

# Rafsun Jany Arman

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## EXPERIENCE

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### Senior Software Engineer

Nov. 2025 – Present

*SELISE Group*

*Dhaka, Bangladesh (Hybrid)*

- Architected a secure, on-premise **Enterprise Generative AI Ecosystem** (Web, Mobile, Backend) using **LangGraph**, **Redis**, and **WebSockets**, optimizing resource utilization to reduce inference latency by ~15% while enabling comprehensive real-time communication.
- Engineered a robust **RAG** pipeline utilizing **Qdrant** and **LangChain** to process complex file structures, delivering context-aware LLM responses with high precision based on private enterprise data.
- Orchestrated a **Multi-Agent System** with **MCP** (Model Context Protocol) tool servers and STT/TTS, integrating specialized agents via **Perplexica** and **SearXNG** for real-time web search and standardized tool access beyond static training data.
- Developed a high-concurrency **Collaborative Chat Engine** with secure link-sharing and dynamic participant management using **FastAPI** and **SignalR**, streamlining group problem-solving.

### Software Engineer

Nov. 2023 – Nov. 2025

*SELISE Group*

*Dhaka, Bangladesh (Remote)*

- Developed the all-in-one **Ski World Cup App** using **Flutter** and **Bitmovin**, delivering sub-3s latency live streaming and real-time athlete telemetry for over 30k users.
- Built a dynamic **Maritime Data Visualization Dashboard** for a global marine services client, enabling predictive alerts and efficient data filtering across multiple operational systems.
- Engineered a scalable **IoT Middleware** using **Java Spring Boot**, **RabbitMQ**, and **TimescaleDB** to orchestrate millions of telemetry messages for Smart Metering and Industrial use cases.
- Designed end-to-end **Smart Metering Integration** solutions, automating data collection from diverse hardware protocols to optimize utility consumption analysis.

### Junior Software Engineer

Aug. 2022 – Oct. 2023

*SELISE Group*

*Dhaka, Bangladesh (Hybrid)*

- Extended an **IoT Commodity Tracking Platform** for a leading mining company, ensuring accurate shipment monitoring and bag condition tracking across international ports.
- Created a **Production Cost Monitoring System** for a Swiss food manufacturer, utilizing IoT data to track electricity usage and streamline operational efficiency, resulting in reduced waste.
- Enabled **White-Label IoT Solutions** by developing customizable branding tools and rule chains, allowing B2B clients to tailor platforms to specific operational branding needs.

### Intern Software Engineer

Mar. 2022 – Jul. 2022

*SELISE Group*

*Dhaka, Bangladesh (Remote)*

- Conducted R&D on **ThingsBoard IoT** platform features and designed custom D3.js data visualization graphs to meet specific client reporting requirements.
- Published an NPM package for file uploads in **NativeScript** (Android API  $\geq 30$ ) and identified security vulnerabilities in internal systems, earning recognition.

## RESEARCH

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### Fish classification using saliency detection | *Computer Vision, U2-Net, HOG*

- Developed a fish classification system utilizing saliency detection (depending on shape and texture) and ensemble learning (KNN, SVM, etc.), achieving up to 100% accuracy on local fish datasets.

### Fish Observatory and Census System (FOCS) | *Image Processing, Research*

- Proposed and implemented a census system for automated fish monitoring and population estimation using advanced image processing techniques.

### Parkinson's Disease Classification | *Deep Learning, DenseNet121, InceptionV3*

- Built an image-based detection system using DenseNet121 and InceptionV3 to classify hand-drawn spirals/waves, achieving high accuracy (96.67%) in differentiating healthy vs. affected subjects.

## Bangla & Bus-Mama | *NLP, Python, Mobile Dev*

- Created ‘Bangla’, a Python library for Bengali text processing and lemmatization using Trie data structures.
- Applied the library in ‘Bus-Mama’, a real-time tracking application with a Bengali-language chatbot for university students.

## PROJECTS

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### CrewsBoard | *Python, CrewAI, FastAPI, React, RAG, Multi-Agent*

- Engineered an AI agent task management microservice (**core**) supporting multi-model architectures, multi-tenancy, and a RAG knowledge base for context retrieval accuracy.
- Built **Canvas**, a modular FastAPI + React + SQLAlchemy UI for orchestrating AI agent workflows with value-based flow management and visualization.

### LiveKit Voice Agent | *Python, LiveKit, LangGraph, Whisper, MCP*

- Built a self-hosted AI voice/video stack with **LangGraph** orchestration, **Whisper** STT, real-time audio handling, and **MCP** tool integration for production agentic voice workflows.

### Adaptive Marketing AI | *Python, LangChain, AI/ML, Marketing API*

- Built an intelligent marketing engine that interfaces with multiple ad platforms to generate and deploy targeted campaigns based on data-driven insights.

### AI Brand Mention Tracker | *TypeScript, Firestore, LLM Observability*

- Built a backend service analyzing how a brand surfaces in AI-generated responses, with scheduler-driven analysis, score-change alerts, and Firestore-based event triggers.

## TECHNICAL SKILLS

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**Languages:** Python, Java, TypeScript, Dart, SQL

**Frameworks:** FastAPI, Spring Boot, React, Flutter (BLoC, Riverpod)

**AI / LLM Stack:** LangGraph, LangChain, CrewAI, Qdrant, MCP (Model Context Protocol), LiveKit, Whisper, Perplexica, SearXNG

**Data & Infra:** PostgreSQL, TimescaleDB, MongoDB, Redis, Firestore, RabbitMQ, Docker

**Concepts:** Production Agentic AI, RAG, Multi-Agent Systems, On-Premise GenAI, Microservices, IoT, System Architecture, CI/CD

## HONORS & AWARDS

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|----------------------------------------------------------------------------------------|------|
| <b>Language &amp; Knowledge Engineering Lab</b>   <i>8th Int. Symposium (LKE 2021)</i> | 2021 |
| <b>Top 100</b>   <i>Idea Contest Mujib 100</i>                                         | 2021 |
| <b>Honorable Mention</b>   <i>CanSat Competition (BRAC)</i>                            | 2019 |
| <b>Trainer</b>   <i>Mechatronics Club, GSTU</i>                                        | 2019 |

## EDUCATION

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**Gopalganj Science and Technology University (GSTU)**

*Bachelor of Science in Computer Science & Engineering; CGPA: 3.41*

Gopalganj, Bangladesh

*Jan. 2017 – May 2022*